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Pepsensyal™

Age slowing peptide

Global Except US

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SUMMARY FILE

Pepsensyal	BC10230
Origin - Description	Age slowing peptide, as acetyl and lauroyl derivatives
Regulatory data	
INCI	Mannitol (and) Acetyl sh-Tetrapeptide-1 (and) Lauroyl sh-Tetrapeptide-1
China	Acetyl sh-Tetrapeptide-1 (and) Lauroyl sh-Tetrapeptide-1 are NOT listed in Inventory of Existing Cosmetic Ingredient in China (IECIC 2021)
CAS#	7732-18-5; 120081-14-3; 2126842-42-8
Appearance	White to off white homogeneous powder
Preservative	None
Natuals labels	None
Naturalness content (ISO 16128)	> 99% from natural origin
Cosmetic use	
Properties	Helps to slow-down cell aging through Sirtuin-1 stimulation (*in vitro*) Helps to boost skin own's regeneration resources through stimulation of growth of keratinocyte stem, progenitor and fibroblast cells (*in vitro*) Helps to upregulate keratinocyte tight junction proteins for epidermal regeneration and cohesion (*in vitro*) Helps to upregulate structural components of DEJ and ECM (GAGs, Collagen III, collagen IV, fibronectin) for dermal regeneration (*ex vivo*) Contributes to dermal density improvement Contributes to improve the appearance of skin surface relief Contributes to reduce the appearance of wrinkles
Applications	Face care, anti-aging, slow-aging, anti-wrinkles
Formulation data	
Concentration of use	0.2%
Solubility	Soluble in water, insoluble in oils and fats
Incorporation method	Pepsensyal can be dissolved in water up to 80°C or at room temperature for cold processing
pH optimal	4-8



Peptides for skin care applications

Consumer Aspiration

Consumers are looking for healthy aging as early as their 20's. They prefer to rely on regime base solutions rather than on procedure-based remedies to address blaring skin concerns. They have an increased awareness of skin health, and desire for cosmetic products delivering potent visible efficacy with safe ingredients.

In the cosmetic industry, skin aging is a persistent and top skin concern. Consumers are really looking for solutions to look younger or delay aging signs. 53% of women still prioritize wrinkles reduction among their TOP 3 skin care concerns and 45% of product launched in 2023 where anti-aging positioned [Source Mintel, 2024]. Traditionally, human has developed a full range of solution to try to mask, correct or erase aging signs on the skin. Nowadays, a new attitude emerges embracing slow-aging concepts. On social media, slow-aging buzzed with +1100% in 2023 compared to 2021. People are now concerned about healthy aging by adopting a holistic preventive skin care approach to embrace numbers instead of fighting them.

Peptides in scope

Peptides are composed of short chains of amino acids. In recent years, peptides have received increased interest in cosmetics, food and various other fields. Their prominent role in nature and their involvement in many physiological processes make them valuable cosmetic ingredients. The high potency, specificity and good safety profile are the main strengths of bioactive peptides. They have various functionalities relevant for skin care, including cell proliferation and migration, anti-inflammatory, extracellular matrix (ECM) protein synthesis, supporting the skin own's functionality to repair itself or regenerate. [Lupo & Cole, 2007; Gorouhi & Maibach, 2009; Zhang & Falla, 2009]. Furthermore, since they are comprised of amino acids, peptides are easily processed by body tissues over time and avoid issues with tolerance [Reddy *et al*, 2012; Fosgerau & Hoffmann, 2015].

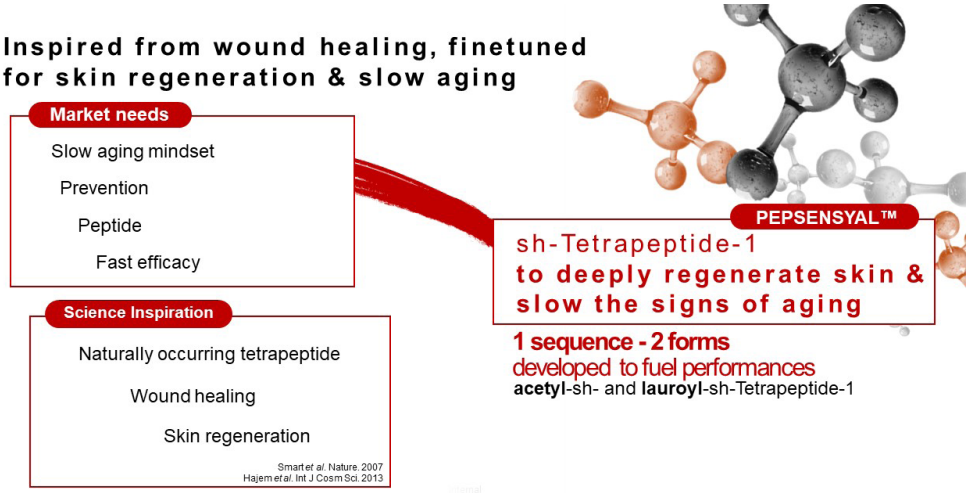
Peptides are now an important class of cosmetic active ingredients and are popular topical agents for alleviating the signs of aging [Gorouhi & Maibach, 2009, Errante *et al*, 2020].

Natural skin regeneration process to reduce and delay the signs of aging

The repair of tissue after injury is a natural process involving skin regeneration mechanisms [Eming *et al*, 2014; Sorg *et al*, 2017]. This process in adult skin is complex, involving several phases, of inflammation, angiogenesis, matrix deposition, cell recruitment and formation of new stratified layer of epidermis leading to the restoration of the skin properties. In healthy skin, aging is also characterized by a progressive alteration of the epidermis and dermis [Robert *et al*, 2012; Tobin, 2017]. Promoting the skin self-regeneration pathways should contribute to provide benefits against these damages, to delay or reduce the signs of the skin aging.

Peptides promoting skin natural continuous regeneration are a field of investigation to provide slow-aging active ingredient for cosmetic applications.

Pepsensyal: a naturally occurring tetrapeptide, inspired by natural skin regeneration process



Acetyl-sh- and Lauroyl-sh-Tetrapeptide-1

Description: Pepsensyal is composed by the acetylated and lauroyl derivatives, on the N-terminal, of sh-Tetrapeptide-1. sh-Tetrapeptide-1 is a single chain human peptide, produced synthetically to be identical to a portion of the protein, Thymosin Beta 4, and contains 4 amino acids consisting of serine, aspartic acid, lysine, proline: Ser-Asp-Lys-Pro (SDKP).

Number of amino acids: 4

Molecular weight: 487.5 and 627.8 Da respectively

Solubility: water soluble

Process: chemical peptide synthesis, then blending with mannitol. All amino-acids are in the natural L-form and not of animal origin.

Inspired by wound-healing research: the acetylated tetrapeptide AcSDKP was identified as a human naturally occurring biopeptide. It is present in the blood at nanomolar concentrations and is formed *in vivo* by the enzymatic cleavage of the N-terminus of thymosin-beta4 [Cavasin *et al*, 2004] which promotes endothelial cell differentiation, has pro-angiogenic effects and accelerates human skin wound healing [Fromes *et al*, 2006, Smart *et al*, 2007]. In an *ex vivo* assay on human skin explants with lesions induced by UVB irradiation, topical AcSDKP application was reported to accelerate healing [Fromes *et al*, 2006].

Safety / tolerability of the product

Pepsensyal was evaluated to ensure its safety in the recommended conditions of use.

Description

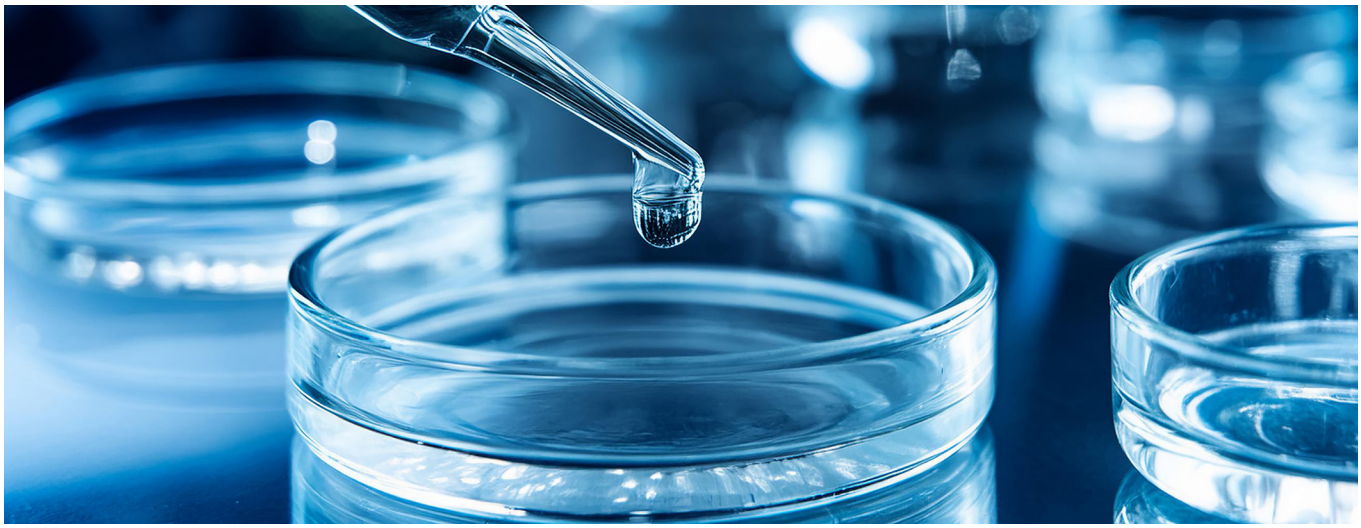
Pepsensyal is a preservative free powder, blending 20-50 ppm of the peptides Acetyl sh-Tetrapeptide-1 and Lauroyl sh-Tetrapeptide-1 with mannitol.

INCI name: Mannitol (and) Acetyl sh-Tetrapeptide-1 (and) Lauroyl sh-Tetrapeptide-1

Recommended dose of use: 0.2%



Example: Illustration of a commercial sample of Pepsensyal incorporated into a hydrogel and a face cream at 0.2%.



Demonstrated efficacy on skin regeneration to slow the signs of aging

The properties of Pepsensyal to promote the own skin regeneration functions and to slow the signs of aging was studied *in vitro* [Hajem *et al*, 2013] and *in vivo* on human volunteers.

In vitro, the properties of Pepsensyal were demonstrated on:

1/ **Slowing cell-aging**:
- by stimulating sirtuin-1 protein expression in human keratinocytes and fibroblasts,
- by stimulating sirtuin-1 enzymatic activity.

2/ **Epidermis regeneration**:
- by stimulating human epidermal keratinocyte stem cell and progenitor cell growth,
- by upregulating keratinocyte tight-junction proteins ZO-1, claudin-1 & occludin.

3/ **Matrix regeneration**:
- by stimulating human dermal fibroblast cell growth,
- by upregulating Dermo Epidermal Junction & Extra Cellular Matrix components in the dermis (*ex vivo*).

In vivo, we studied in a placebo-controlled clinical study the effects of Pepsensyal at 0.2% for slow aging on facial skin density, surface structure and wrinkles appearance.

EFFICACY

Slowing cell-aging: boosting Sirtuin-1 protein expression and activity for cell longevity

in vitro

OBJECTIVE

Sirtuin-1 (SIRT1) is a member of a highly conserved gene family (sirtuins) encoding NAD^+-dependent deacetylases, originally found to deacetylate histones, leading to increased DNA stability and prolonged survival in mammal cells. Sirtuin is an essential factor that delays cellular senescence and extends the cellular lifespan [Mouchiroud *et al*, 2013; Houtkooper *et al*, 2016; Grabowska *et al*, 2017; Lee *et al*, 2019]. SIRT1 is functionally expressed in human-skin keratinocytes and fibroblasts [Dal Farra & Domloge, 2006].

The aim of the study was to evaluate *in vitro* the stimulatory effect of Acetyl Sh-Tetrapeptide-1 from Pepsensyal on the Sirtuin-1 expression on human keratinocytes and fibroblasts, and on the Sirtuin-1 activity [Hajem *et al*, 2013].

RESULTS & DISCUSSION

Stimulation of Sirtuin-1 protein expression [Hajem *et al*, 2013]

The effect of Acetyl sh-Tetrapeptide-1 on the expression of Sirtuin-1 protein was examined *in vitro* using human keratinocytes (HaCaT) and dermal fibroblasts (NDHF) cells. Sirtuin-1 expression was investigated by immunostaining (Fig. 1) and quantified by Western blotting (Fig. 1) on keratinocytes. These assays with 0.01 to 100 nM of Acetyl sh-Tetrapeptide-1 were conducted using Resveratrol (10 µM), a well-known stimulator of Sirtuin-1 [Rogina & Tissenbaum; 2024], as positive control.

Resveratrol was reported to increase Sirtuin-1 protein expression in human keratinocytes [Lee *et al*, 2016]. SIRT1 is downregulated by both UV exposure and H_2O_2, two inducers of skin cell DNA damage and aging [Cao *et al*, 2009]. These DNA damages were prevented by resveratrol, suggesting that SIRT1 stimulators may protect skin against the adverse effects of solar UV radiation and oxidative stress. Moreover, it was reported that enforced SIRT1 expression promoted cell proliferation and antagonized cellular senescence in human fibroblasts [Huang *et al*, 2008].

Pepsensyal's Acetyl sh-Tetrapeptide-1 upregulated the synthesis of Sirtuin-1 in both types of skin cells. The results of Western blot analysis presented in Figure 2 showed an increase at the 3 different concentrations tested 0.01 nM, 1 nM and 100 nM, by +70%, +180% and +170% respectively.

Interestingly, compared to the reference Resveratrol, Acetyl sh-Tetrapeptide-1 has similar activity at a 10.000x lower dose (1 nM *vs* 10 µM).

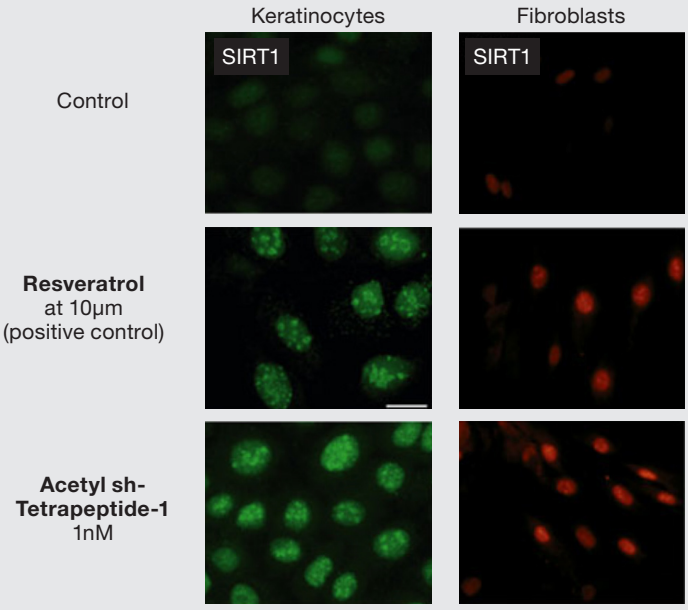
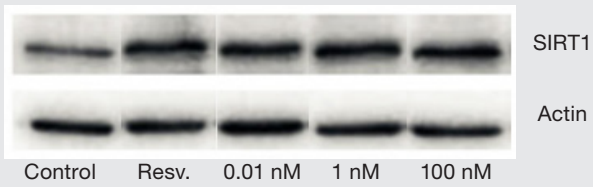


Figure 1 - Immunocytochemical staining of Sirtuin-1 in human epidermal HaCat keratinocytes (green-labelled) and human dermal fibroblasts (red-labelled). 1 nM Acetyl sh-Tetrapeptide-1 is equivalent to ~0.002% Pepsensyal – Scale bar 20 µM.

in vitro



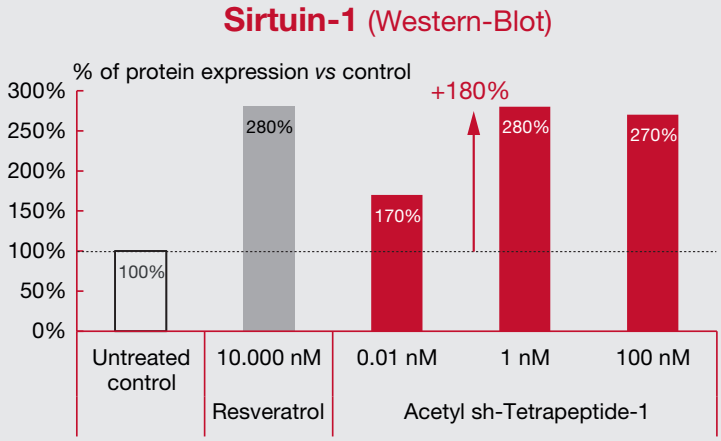


Figure 2 - Western blot analysis by densitometric analysis of Sirtuin-1 (SIRT1) level in HaCaT cells after 24h treatment with Acetyl sh-Tetrapeptide-1 at different doses or Resveratrol a 10.000 nM (10 µM). Actin was used as a loading control. 1 nM Acetyl sh-Tetrapeptide-1 is equivalent to ~0.002% Pepsensyal.

Stimulation of Sirtuin-1 enzymatic deacetylase activity, *in tubo* assay

The direct effect of Acetyl sh-Tetrapeptide-1 on enzymatic activity of SIRT1 was examined *in tubo*. SIRT1 activity was measured by a deacetylation assay using as a substrate a fluorogenic acetylated peptide. An increase in SIRT1 activity at concentrations of 0.01 nM to 100 nM, was observed, reaching +40% for 1 nM (Fig. 3).

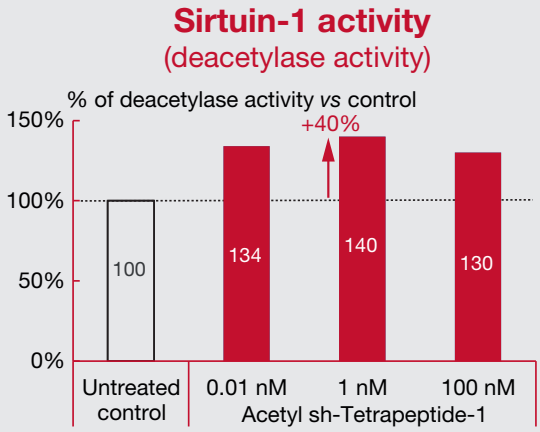


Figure 3 - Sirtuin-1 enzymatic deacetylase activity, *in tubo* assay. 1 nM Acetyl sh-Tetrapeptide-1 is equivalent to ~0.002% Pepsensyal.

CONCLUSION

At low dose, in nanomolar range, Pepsensyal's Acetyl sh-Tetrapeptide-1 boosts Sirtuin-1 protein expression by skin cells and its enzymatic deacetylase activity *in vitro*, and thus may contribute to extend cellular lifespan and to slow-down skin cell aging.

MATERIALS&METHODS

Sirtuin-1 (SIRT1) protein expression *in vitro*

Human keratinocytes (HaCaT) or normal human dermal fibroblasts (NHDF) were seeded in six-well plates at a density of $2x10^5$ cells/mL with complete culture medium. After 24h of culture, the untreated control and Acetyl sh-Tetrapeptide-1 treated cells were rinsed with cold PBS and protein were extracted with cell lysis buffer (Cell Signaling Technology, Danvers, MA, U.S.A.) supplemented with protease inhibitors (Sigma). Protein (20 µg) were resolved in 10% SDS-PAGE and transferred onto nitrocellulose membrane (Millipore). After blocking with 5% w/v non-fat dry milk in TBS, membranes were incubated overnight at 4°C with anti-SIRT1 primary antibodies (Santa Cruz Biotechnology), followed by anti-IgG conjugated to horseradish peroxidase (Life Technologies). Blots were reprobed with antibody against β-actin (Santa Cruz Biotechnology) for normalization. Enhanced chemiluminescence was used for detection (Millipore) following manufacturer's instructions. Immunoreactive proteins were visualized with ChemiDoc XRS (BioRad, Marnes-la-Coquette, France) and densitometric analysis was performed with QUANTITY ONE 1-D analysis software (Bio-Rad).

Sirtuin-1 (SIRT1) deacetylase activity *in tubo*

SIRT1 deacetylase activity was determined using an HTRF SIRT1 activity assay kit (Cisbio International, Saclay, France) according to the manufacturer's recommendations. Briefly, the HTRF deacetylation assay is performed in two steps: an activation step in presence of enzyme and acetylated substrate, and a revelation step in which the deacetylation event is quantified. The maximum signal obtained in the absence of enzyme decreases when the deacetylation reaction occurs. The quantity of deacetylated peptides is then directly correlated to the signal extinction. Deacetylation assay was carried out in low volume 384-well plates. Enzymatic reaction was performed in 10 µl at room temperature for 1 hour in presence of 150 µM of NAD^+, 6 mU of SIRT1 (Enzo Life Sciences, Villeurbanne, France) and 6 nM of acetylated substrate without (untreated control) or with Acetyl sh-Tetrapeptide-1. The reaction was stopped by adding 10 µl of anti-acetylated substrate monoclonal antibody labeled with $Eu3^+$ cryptate diluted in a buffer containing EDTA and SIRT1 inhibitor (nicotinamide). After overnight incubation, the cryptate is excited at 337 nm and the fluorescence is measured at 620 nm (cryptate emission wavelength) and 665 nm (d2 emission wavelength) using a microtiter plate reader POLARstar OMEGA (BMG LabTech, Champigny-sur-Marne, France). Deacetylation of substrate is calculated as followed: 100- [(ratio sample/ ratio no enzyme) x 100] with ratio: (665 nm / 620 nm) $x10^4$.

EFFICACY

Epidermis regeneration: boost epidermal stem cells for continuous epidermis regeneration

in vitro

OBJECTIVE

The skin constantly renews itself throughout adult life. Stem cells residing in the basal layer of the epidermis (interfollicular epidermal stem cells) exhibit the essential ability for self-renewal and differentiation [Abbas & Mahalingam, 2009; Mascré *et al*, 2012; Tang *et al*, 2024]. They give rise to daughter transient amplifying progenitor cells which proliferate, differentiate and are essential for wound repair and for maintaining adult-skin homeostasis (Fig 4).

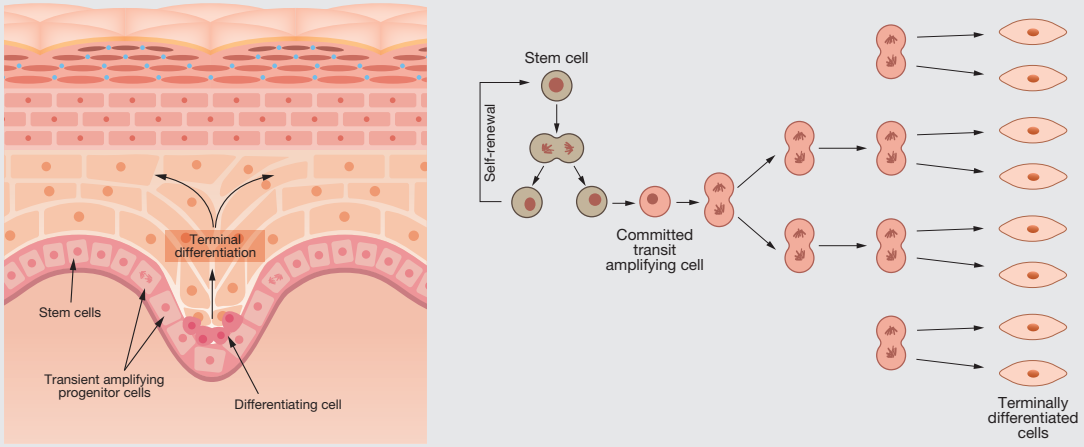


Figure 4 - Schematic drawing illustrating epidermal stem and transient amplifying progenitor cells in the epidermis. Source: modified from Alberts *et al*, 2008.

However, when they are aging their potency to restore the senescent cell in the skin is compromised [Abbas & Mahalingam, 2009]. It was already reported, *in vitro*, that chronological ageing induced loss of stem cells proportion in the skin [Youn *et al*, 2004]. Chronological or premature aging of skin may be associated with dysfunctions or loss of skin-resident stem/progenitor cells, which thus may impair the skin regeneration process and wound healing [Mimeault & Batra, 2010]. Aging-induced changes in gene expression affect epidermal stem cells interactions with their niches, altering division and motility, with a slowing down of the proliferative rate of keratinocytes in the basal layer contributing to epidermal atrophy and fragility [Tang *et al*, 2024].

Even if they are some debates on whether a reduction in epidermal stem cells proliferative capacity contributes to skin aging [Giangreco *et al*, 2008; Tang *et al*, 2024], the search for compounds that are able to ensure proliferation and functionality of epidermal stem cells is one of the areas of research for skin regenerative active ingredients [Tan *et al*, 2019].

The aim of the study was to evaluate *in vitro* the stimulatory effect of Acetyl sh-Tetrapeptide-1 from Pepsensyal on the growth capacity of epidermal stem cells and transient amplifying progenitor cells in 2D cultures [Hajem *et al*, 2013].

RESULTS & DISCUSSION

Upregulation of the growth of human keratinocyte stem and progenitor cells

This study has been performed with clonogenic assays on human epidermal keratinocytes of neonatal foreskin sample. A clonogenic assay (or colony formation assay) is an *in vitro* cell survival assay based on the ability of a single cell to grow into a colony.

The basal keratinocyte cell subpopulations were sorted by flow cytometry combining an adhesion molecule-marker (integrin-α6) with a proliferation associated marker (transferrin receptor, Trf-R) into quiescent keratinocyte stem cells and cycling transient amplifying progenitor cells. The cells were then treated or not (untreated control) for 14 days before determining the ratio of the number of colonies to the number of seeded cells.

in vitro

Figure 5 shows a statistically significant increase in the number of keratinocyte stem cells (+35% and +22%, p<0.001) and progenitors (+40%, p<0.001, and +22%, p<0.01) colonies following treatment with Acetyl sh-Tetrapeptide-1 tested at 0.1 nM and 10µM respectively. Interestingly, this effect of +35% on stem cells and +40% on progenitor cells was more important for the lower concentration of the peptide, 0.1 nM.

These results provide evidence of a growth stimulatory effect of Acetyl sh-Tetrapeptide-1 on human epidermal keratinocyte stem cells and progenitor cells, and that it might help maintain a higher number of stem / progenitor cells in the epidermis over time to ensure the permanent renewal of the epidermis.

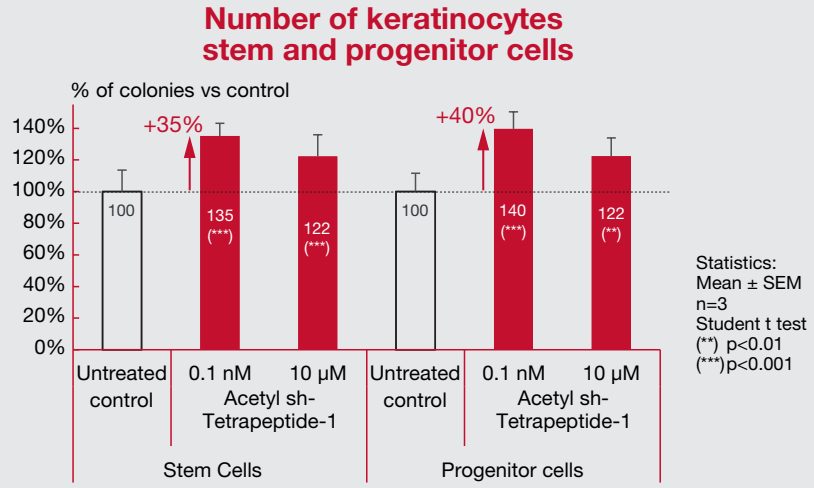


Figure 5 - Clonogenic assay (or colony formation assay) on quiescent keratinocyte stem cells and cycling transient amplifying progenitor cells treated or not with Acetyl sh-Tetrapeptide-1. 0.1 nM Acetyl sh-Tetrapeptide-1 equivalent to ~0.0002% Pepsensyal.

CONCLUSION

At low dose, in nanomolar range, Pepsensyal's Acetyl sh-Tetrapeptide-1 boosts keratinocyte stem and progenitor cells *in vitro*, and thus could help the epidermis to boost its own regeneration resources to delay skin aging.

MATERIALS&METHODS

Human keratinocyte cells

Experiments of clonogenic survival assays have been performed using human interfollicular epidermal keratinocytes of neonatal foreskin samples from routine circumcisions and obtained after informed consent of the patient's parents. Human neonatal biopsies were incubated overnight at 4°C in a solution containing DMEM, dispase 2.4 U/mL, trypsin 0.5 g/L and 2.5% antibiotics from Life Technologies (Saint-Aubin, France). Epidermal sheets were then separated from the dermis. The epidermis was incubated in trypsin-EDTA for 20 min at 37°C. Cell dissociation was performed by gentle pipetting and filtering on 70 µm cell strainers (BD Biosciences, Le Pont-de-Claix, France) to obtain a single-cell suspension. Then, keratinocytes were centrifuged at 200 g for 15 min and cell viability was assessed using trypan blue (Sigma, Saint-Quentin Fallavier, France).

Cell staining and sorting

Basal keratinocyte subpopulations were freshly isolated from tissue samples based on two cell-surface markers, an adhesion molecule (integrin-α6) combined with a proliferation-associated marker (transferrin receptor, Trf-R) according to Kaur's laboratory protocol [Li *et al*, 1998]. Cell immunostaining was performed in PBS-2% BSA (Sigma) and required an incubation with c-globulins (Beckman Coulter, Villepinte, France), followed by staining with R-phycoerythrin-conjugated rat anti-human CD49f/integrinα6 (Itg-α6) antibody (clone GoH3; BD Biosciences) and streptavidin allophycocyanin mouse anti-human CD71/transferrin receptor (Trf-R) antibody (BD Biosciences). Then, cell populations were sorted by flow cytometry (MoFlo; Dako, Trappes, France). Trf-R expression does successfully separate the integrin-α6 bright (Itg-$\alpha6^{bri}$) proliferative compartment of human epidermis into quiescent [keratinocyte stem cells (KSC)] [itg-$\alpha6^{bri}$ $CD71^{dim}$] and cycling [transient amplifying progenitor cells (TA)] [itg-$\alpha6^{bri}$ $CD71^{bri}$] tissue reconstituting cells.

Clonogenic survival assay

Human primary sorted basal keratinocytes (KSC and TA) were seeded at a clonal density of 5 keratinocytes/cm^2 onto lethally irradiated human fibroblasts in Green L1 medium. This culture medium contained DMEM and Ham's F12 media (3 : 1 mixture; Life Technologies), 10% FCS (Thermo Fisher Scientific, Villebon-sur-Yvette, France), 10 ng/mL epidermal growth factor (Millipore SAS, Molsheim, France), 5 µg/mL transferrin, 5 µg/mL insulin, 0.4 µg/mL hydrocortisone, 180 µM adenine, 2 mM tri-iodothyronine (Sigma), 2 mM L-glutamine (Life Technologies), and 100 U/mL penicillin/streptomycin (Life Technologies). Medium was renewed every 2 days. Cells were allowed to attach to culture dishes overnight at 37°C in a humidified atmosphere with 5% CO_2. After 24 h, 100 µL of Acetyl sh-Tetrapeptide-1 solution in water was added or not (untreated control) to the culture medium at two final concentrations (10 µM and 0.1 nM). Individual colonies were quantified after 2 weeks of culture. Colony staining was performed with eosin and blue reagent (RAL 555) from RAL Diagnostic (Martillac, France). Keratinocyte colony-forming efficiency corresponds to the ratio of the number of colonies to the number of seeded cells. The % of colony were determined at each concentration and compared with the control vehicle-treated cells. Three experiments were carried out using independent foreskin samples and statistical comparisons were performed using the Student t-test. Data are presented as mean percentage ± SEM values, *vs* the untreated control as base 100%, and $p < 0.05$ was considered statistically significant.

EFFICACY

Epidermis regeneration: upregulates keratinocyte tight junction proteins for epidermal cohesion

in vitro

OBJECTIVE

The cohesion of epidermal cells is a fundamental to keep a healthy epidermis. Reinforcing joinery between cells in epidermis through tight junction (TJ) proteins upregulation is key for the epidermal cohesion [Brandner *et al*, 2002].

TJs are cell-cell adhesive zipper-like structures which ensure cell-cell cohesion in the epidermis, prevent diffusion of solutes through the intercellular spaces and protect the skin from unnecessary water loss as well as the entry of detrimental influences from the outside [Morita & Miyachi, 2003]. A variety of TJ proteins including claudins (such as claudin-1), occludin, zonula occludens-proteins (ZO, such as ZO-1) have been identified in mammalian epidermis [Kirschner & Brandner, 2012]. In aged human skin, there is a decrease in the expression of the tight junction components claudin-1 and occludin [Jin *et al*, 2016], skin barrier is impaired and following insult there is decreased recovery of function [Parish, 2017].

The aim of the study was to evaluate *in vitro* the stimulatory effect of Acetyl sh-Tetrapeptide-1 from Pepsensyal on the expression of three structural components of tight junctions, ZO-1, claudin-1 and occludin, in human keratinocytes [Hajem *et al*, 2013].

RESULTS & DISCUSSION

Upregulation of tight junction proteins in human keratinocytes

The expression of three major structural components of tight junctions (TJ) in the human keratinocytes was firstly evaluated by immunocytochemistry (Fig. 6). When treated with Acetyl-sh-Tetrapeptide-1 at 1nM, increases in the expression of ZO-1, claudin-1 and occludin were visualized in human HaCaT keratinocytes.

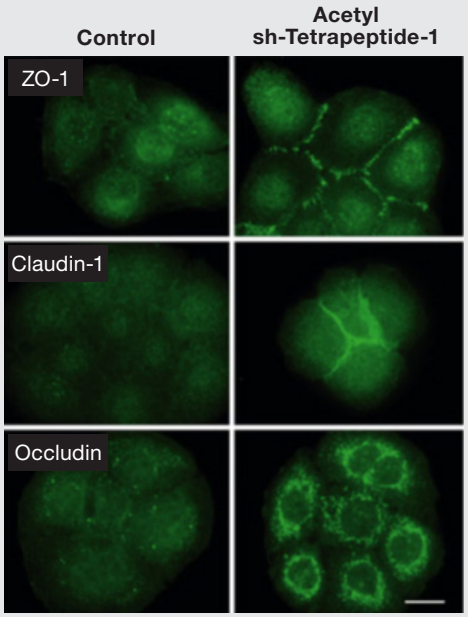


Figure 6 - Immunocytochemical staining of ZO-1, claudin-1 and occludin in human epidermal HaCaT keratinocytes (green-labelled). Scale bar 20 µM. 1 nM Acetyl sh-Tetrapeptide-1 is equivalent to ~0.002% Pepsensyal.

These data were next validated by Western blot analysis on HaCaT keratinocyte cell lysates. As shown in figures 7A and B, respectively for the doses of 0.01 nM, 1nM and 100 nM of Acetyl sh-Tetrapeptide-1, ZO-1 and claudin-1 expression were increased respectively in a dose dependent manner by +7%, +41% and +66% and by +23%, +46% and +57%. Occludin expression followed a dose-inverse response with an increase by +62%, +37% and +10% at the same doses.

These findings show that the expression of all the 3 studied tight junction proteins were increased at 1 nM dose, respectively by +41%, +46% and +37% for ZO-1, claudin-1 and occludin, and suggest that this peptide may lead to better epidermis cohesion and control of skin permeability.

in vitro

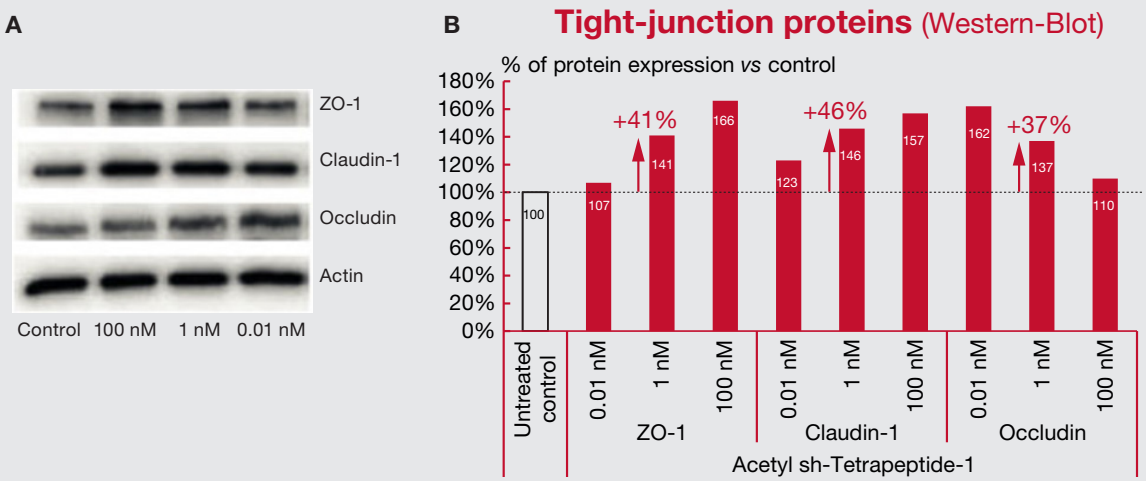


Figure 7 - **A**- Western blot analysis of ZO-1, claudin-1 and occludin from cell lysates in HaCaT keratinocytes. Actin was used as a loading control. **B**- Relative expression ratio, differences in loading were adjusted to actin protein level. 1 nM Acetyl sh-Tetrapeptide-1 is equivalent to ~0.002% Pepsensyal.

CONCLUSION

At low dose, in nanomolar range, Pepsensyal's Acetyl sh-Tetrapeptide-1 upregulates keratinocyte tight junction proteins *in vitro*, and thus could help the epidermis remodeling for better epidermis cohesion, control of skin permeability and delay of skin aging.

MATERIALS&METHODS

Human keratinocytes (HaCaT) or normal human dermal fibroblasts (NHDF) were seeded in six-well plates at a density of $2x10^5$ cells/mL with complete culture medium. After 24h of culture, the untreated control and Acetyl sh-Tetrapeptide-1 treated cells were rinsed with cold PBS and protein were extracted with cell lysis buffer (Cell Signaling Technology, Danvers, MA, U.S.A.) supplemented with protease inhibitors (Sigma). Protein (20 µg) were resolved in 10% SDS-PAGE and transferred onto nitrocellulose membrane (Millipore). After blocking with 5% w/v non-fat dry milk in TBS, membranes were incubated overnight at 4°C with anti-claudin-1, anti-occludin and anti-ZO-1 primary antibodies (Zymed Laboratories) followed by anti-IgG conjugated to horseradish peroxidase (Life Technologies). Blots were reprobed with antibody against β-actin (Santa Cruz Biotechnology) for normalization. Enhanced chemiluminescence was used for detection (Millipore) following manufacturer's instructions. Immunoreactive proteins were visualized with ChemiDoc XRS (BioRad, Marnes-la-Coquette, France) and densitometric analysis was performed with QUANTITY ONE® 1-D analysis software (Bio-Rad).

EFFICACY

Matrix regeneration: upregulates Dermo Epidermal Junction & Extra Cellular Matrix components

ex vivo

OBJECTIVE

Skin aging is characterized by a progressive deterioration of its functional properties, linked to alterations of the dermo epidermal junction (DEJ) linking the epidermis and dermis, and of the dermal extracellular matrix (ECM) [Robert *et al*, 2012; Tobin, 2017].

The aim of the study was to evaluate *ex vivo* the stimulatory effect of Acetyl sh-Tetrapeptide-1 from Pepsensyal on the level of structural components of DEJ and ECM in human skin explants [Hajem *et al*, 2013].

RESULTS & DISCUSSION

Upregulation of DEJ and ECM components in human skin explants

As shown in the figure 8 pictures, the topical application on skin explants of Acetyl sh-Tetrapeptide-1 increased, after 6 days, the amount of Glycosaminoglycans (GAGs), collagen III, IV and fibronectin at the DEJ, as compared to the control treated with saline solution. GAGs, collagen III and fibronectin were also increased in the papillary dermis. All of these effects were noted with both 10 nM (equivalent to ~0.02% of Pepsensyal) and 10 µM of the Acetyl sh-Tetrapeptide-1 topical applications, even if the efficacy with the lower dose was slightly weaker (data not shown).

These molecular modifications may contribute to a restructuring, regenerating and strengthening effect of the DEJ and of the dermis.

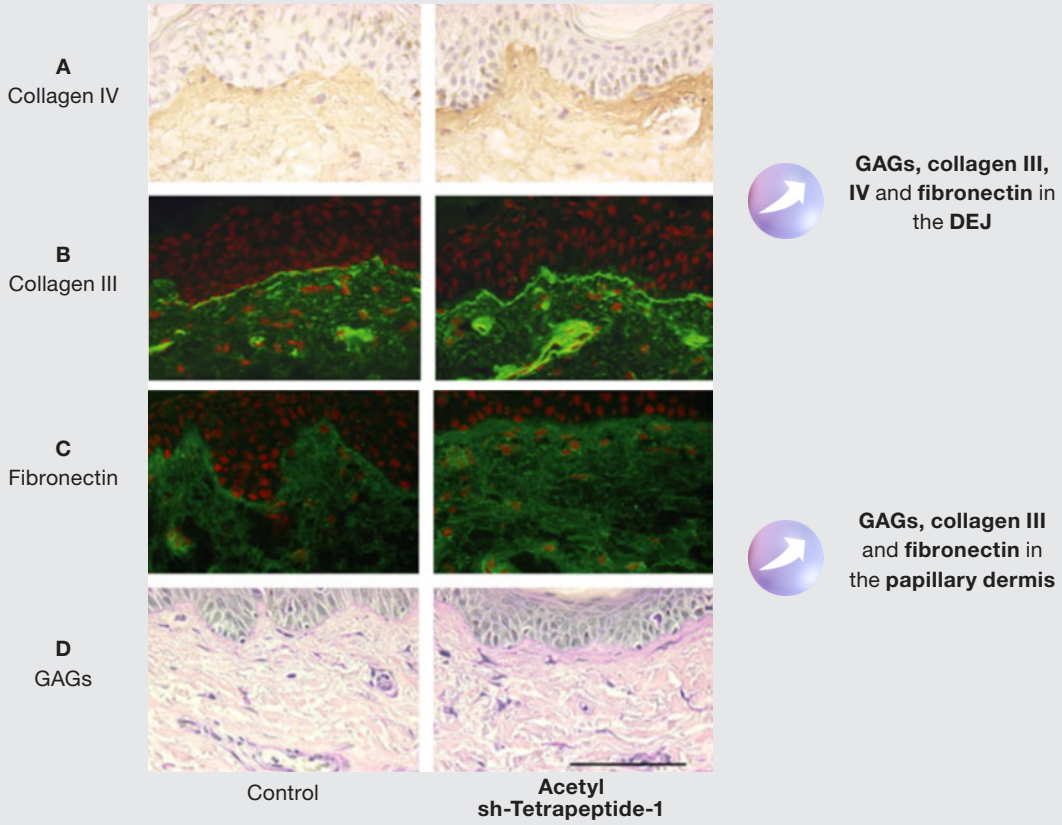


Figure 8 - On skin explants *ex vivo*, application for 6 days every 2 days of Acetyl sh-Tetrapeptide-1 in saline to the explant upper surface, *vs* untreated control. **A**- collagen IV immunolabeling (in yellow); **B**- collagen III immunolabeling (in green); **C**- fibronectin immunolabeling (in green); **D**- GAGs staining with combined alcian blue-PAS method (in pink/violet). Nuclei were stained either with hematoxylin or propidium iodide. Scale bar = 100 µM.

CONCLUSION

At low dose, in nanomolar range, Pepsensyal's Acetyl sh-Tetrapeptide-1 upregulates structural components of DEJ and ECM (GAGs, Collagen III, collagen IV, fibronectin) *ex vivo*, and thus could help restructuring, regenerating and strengthening the DEJ and of the dermis to delay skin aging.

MATERIALS&METHODS

Ex vivo human skin-explant assay

The human skin explants were obtained after informed consent from healthy patients undergoing corrective plastic surgery. Subcutaneous fat tissue was removed, and skin tissue was washed, cut into round 1 cm^2 sections and transferred in six-well plates. The explants were placed epidermal surface upwards on a stainless-steel grill allowing the immersion of their lower surface in skin survival culture medium (Bio-Ec Laboratory, Longjumeau, France). All skin explants were cultured at 37°C in a humidified atmosphere containing 5% CO_2. Half of the medium was replaced every 2 days. Acetyl sh-Tetrapeptide-1 solution in saline (10 nM and 10 µM) was applied to the explant's dry upper surface every 2 days. The controls were treated with saline. After 6 days, three explants for each batch were harvested for histological analysis. Each skin explant was divided into two parts, one part was fixed in standard Bouin solution for 48 h, dehydrated and embedded in paraffin, the second one was frozen in liquid nitrogen and stored at -80°C for later processing.

Histology and immunohistochemistry

To visualize general skin morphology, 5 µm sections of paraffin-embedded tissues were deparaffinized, stained using Masson's trichrome (Labonord S.A., Templemars, France) and analysed by light microscopy. Cryostat sections (7 µm thick) of frozen skin explants were fixed in acetone and then analyzed for the expression of skin macromolecular components using immunochemistry. The antibodies used in the study included anti-fibronectin (clone TV-1; Euromedex, Mundolsheim, France), polyclonal anti-collagen III (Tebu-Bio) and polyclonal anti-collagen IV (CliniSciences, Nanterre, France). Glycosaminoglycans (GAGs) were stained according to combined alcian blue-PAS method. The biotinylated secondary antibodies (Amersham Biosciences, Saclay, France) were revealed either with streptavidin-FITC or streptavidin-alkaline phosphatase from TebuBio. Nuclei were labelled either using hematoxylin or propidium iodide (Sigma). Images were taken on a DLMB microscope (Leica Microsystems SAS, Rueil-Malmaison, France).

EFFICACY

Improvement of skin aging signs from inside to the skin surface: dermis density, skin isotropy and wrinkles

in vivo

OBJECTIVE

Among the many aspects that contribute to the wellness of individuals, healthy and younger-looking skin play a relevant role [Forte *et al*, 2015]. The aging process can cause changes in many parts of the face with loss of skin density, modification of surface relief with decreased isotropy, and wrinkles [Lagarde *et al*, 2005; Robert *et al*, 2012].

In a hemi-face, double-blind, placebo controlled clinical study on volunteers in France, we evaluated the effects of Pepsensyal at 0.2% for its ability to improve the above aging signs on the face (Fig. 9).

32 female volunteers with Fitzpatrick skin type I-IV,
aged 44-64 years old (mean 54.7),
with dry skin on the face, wrinkles and fine lines on the Crow's feet,
and loss of firmness on the lower face.
Randomized split face, twice daily self-applications for 28 days.



Evaluation of dermis density by echography (DermaScan C)
on the lower face, of skin relief isotropy by fringe projection (Primos)
on the cheek, and of wrinkles by fringe projection (Primos)
on the crow's feet.
Time points: D0, D14, D28

Figure 9 - Clinical study design.

The principles of the methods used for this evaluation are briefly explained below.

Evaluation of dermis density by echography (Fig. 10)

The skin aging processes affect the activities related to synthesis, remodeling, and catabolism of the extracellular matrix (ECM) structural components of the dermis [Quan and Fisher, 2015]. As a result, the aging process induce a reduction of the ECM density in the dermis [Robert *et al*, 2012; Tobin, 2017]. Echography enables to evaluate this dermis density through echogenicity measurement [Crisan *et al*, 2012]. An increase of echogenic proportion, *i.e.* more pixels, indicates a densifying effect.

Echogenicity (pixels) of the dermis is
a marker of collagen density in the
extracellular matrix



Low echogenic zone (less pixels)
corresponds to damaged collagen and loss
of structural fibers in the upper dermis

Figure 10 - Skin dermis density, echogenicity and aging.

in vivo

Skin relief isotropy by fringe projection (Fig. 11)

Skin cutaneous lines are signs of the skin ability to stretch and contract in different directions

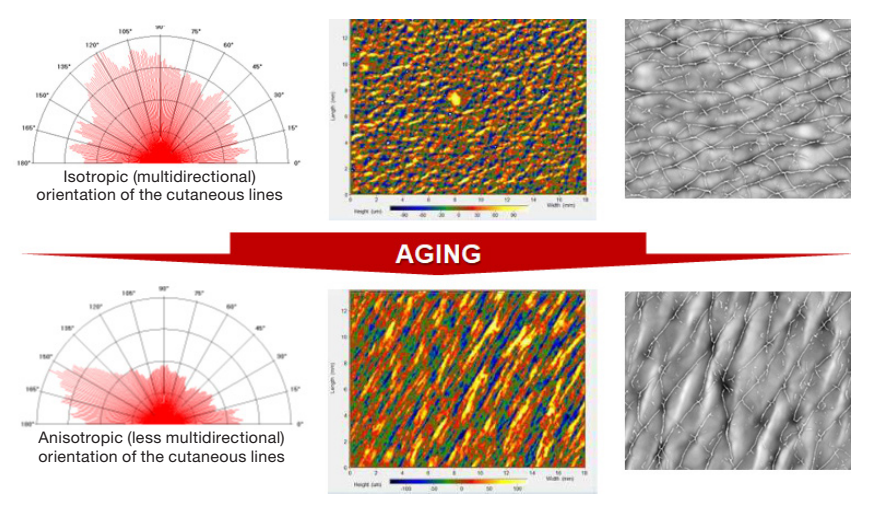


Figure 11 - Skin cutaneous relief, isotropy and aging.

Microscopic observations of human skin relief revealed that skin topography has a network of lines. These lines reflect the underlying mechanical condition of the skin and correspond to the distribution of tension forces in the skin, *i.e.* in the collagen and elastic fibers, as well as the extracellular matrix, and are signs of the skin ability to stretch and contract in different directions [Lagarde *et al*, 2005; Pawlaczyk *et al*, 2013; Thieulin *et al*, 2020]. A transition between a relative isotropic to a highly anisotropic orientation of the cutaneous lines has been observed as a function of aging, *i.e.* the skin ability to stretch and contract is less multidirectional than in young skin (in other words, line directions are more concentrated). Fringe projection enables to determine the isotropy of the skin surface relief. An increase of Isotropy (more multidirectional lines), synonym of a decrease of anisotropy, shows a restructuring effect on the skin surface relief.

Skin roughness and wrinkles by fringe projection

One of the major age-related changes in the skin is the development of wrinkles, particularly on the face which is subject to extrinsic aging factors such as UV exposure [Tobin, 2017; Robert *et al*, 2012]. Fringe projection enables to determine skin roughness and wrinkles parameters on the crow's feet area: the average roughness (Ra), the average relief (Rz) and maximum relief (Rt). A decrease of average roughness indicates a smoothing effect. A decrease of average and maximum relief indicates an anti-wrinkle effect.

RESULTS & DISCUSSION

Effect on dermal density

The 0.2% Pepsensyal serum showed a statistically significant improvement of the dermal density (increase of echogenic proportion) compared to the baseline at both D14 and D28, respectively by +12% ($p<0.01$) and +37% ($p<0.001$) (Fig. 12). At the end of the study, after 28 days of use, the difference with the placebo reached statistical significance, with an increase of the dermal density by 21% ($p<0.01$) comparatively to the placebo.

This result shows that the skin regenerating properties of Pepsensyal contributes to replenish the skin from inside through the improvement of dermis density, with a statistically significant difference *vs* placebo after 28 days of use.

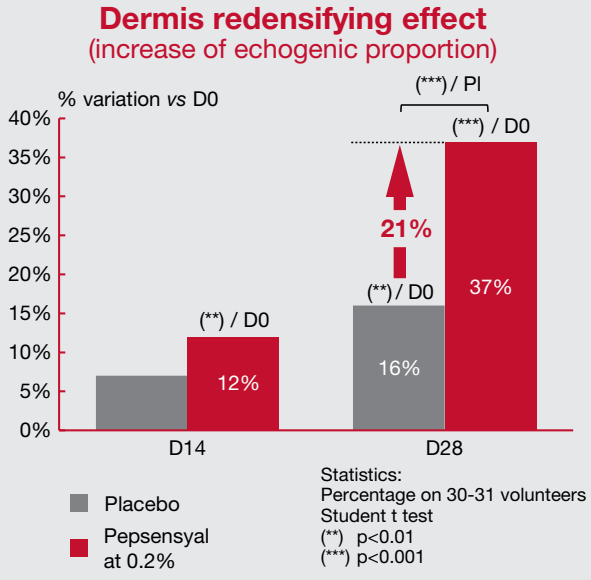


Figure 12 - Mean percentage of change of the dermis density (increase of echogenic proportion) after application of 0.2% Pepsensyal serum *vs* the placebo serum.

in vivo

Pictures in figure 13 illustrate this dermis densifying effect with Pepsensyal, overtime and comparatively to placebo.

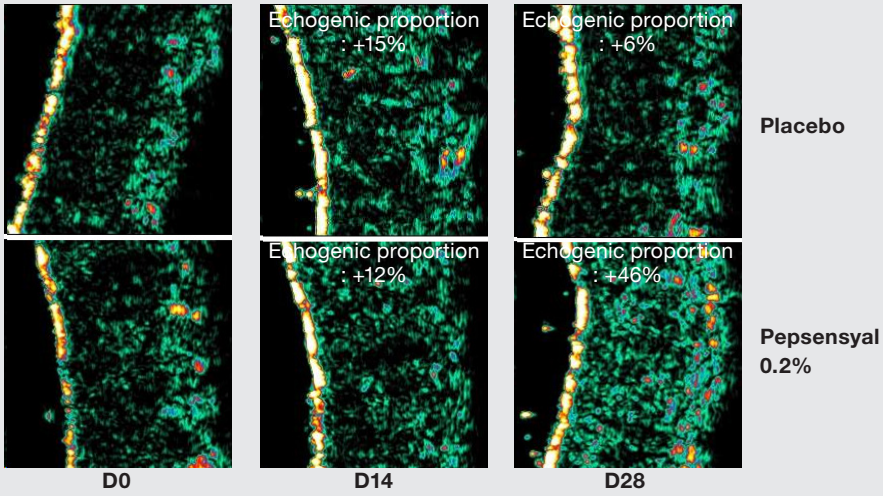


Figure 13 - Illustrative pictures of dermis echogenicity after use of Pepsensyal at 0.2% or placebo, at D0 (baseline, before treatment), and after 14 and 28 days.

Skin surface relief restructuring effect: isotropy

The 0.2% Pepsensyal serum showed a statistically significant increase of the isotropy of the cutaneous relief compared to the baseline and placebo, at both D14 and D28. After only 14 days of Pepsensyal use, isotropy increased respectively by +12% ($p<0.001$) and +15% ($p<0.01$) *vs* baseline and placebo (Fig. 14). At the end of the study, after 28 days of use, this effect was confirmed with respectively +14% ($p<0.01$) and +10% ($p<0.05$) *vs* baseline and placebo.

This result shows that the skin regenerating properties of Pepsensyal contributes to restructure the skin surface through the improvement of cutaneous relief isotropy, with a statistically significant difference *vs* the placebo after only 14 days of use.

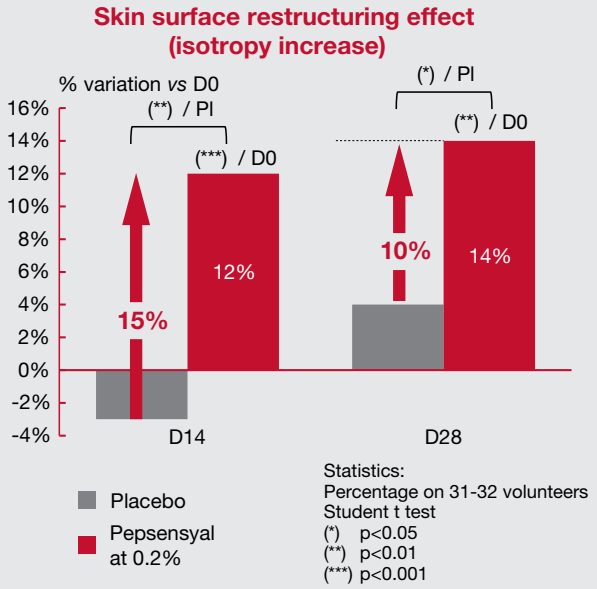


Figure 14 - Mean percentage of change of the skin isotropy after application of the 0.2% Pepsensyal serum *vs* the placebo serum.

Pictures in figure 15 illustrate the modification of cutaneous line isotropy with Pepsensyal, overtime and comparatively to placebo.

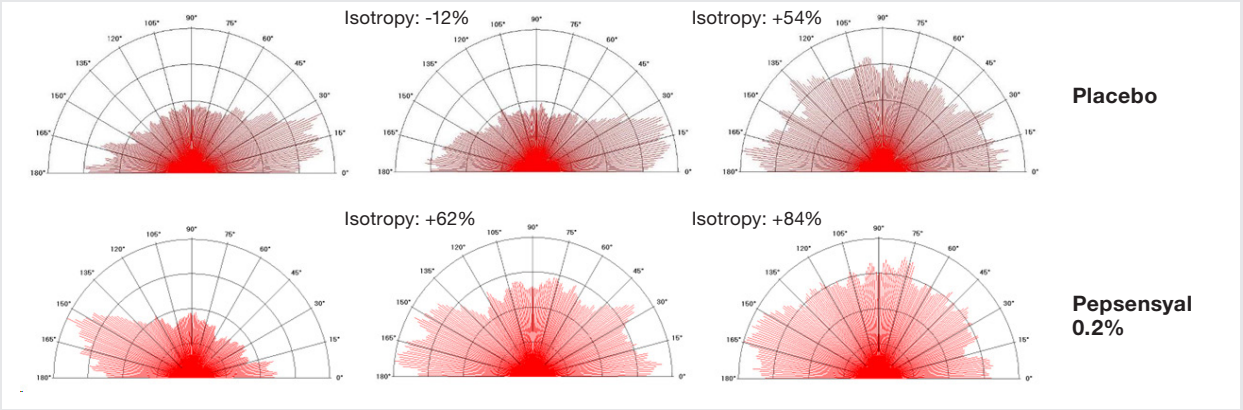


Figure 15 - Illustrative pictures of the skin isotropy with Pepsensyal at 0.2% and placebo, at D0 (baseline, before treatment), and after 14 and 28 days of product use.

in vivo

Skin smoothing and anti-wrinkle effect on the crow's feet area

The 0.2% Pepsensyal serum showed a statistically significant increase of the average roughness (Ra) compared to the placebo, at both D14 and D28 (Fig. 16). After only 14 days of Pepsensyal use, skin roughness decreased by -11% ($p<0.01$) *vs* placebo. At the end of the study, after 28 days of use, this effect was confirmed with -10% ($p<0.05$) *vs* placebo.

On the same manner, average relief (Rz) and maximum relief amplitude (Rt) were significantly decreased compared to the placebo, at both D14 and D28 (Fig. 16). After only 14 days of Pepsensyal use, Rz and Rt were respectively decreased by -12% ($p<0.001$) and -14% ($p<0.001$) *vs* placebo. At the end of the study, after 28 days of use, these effects were confirmed with respectively -12% ($p<0.001$) and -16% ($p<0.01$) for Rz and Rt *vs* placebo.

These results shows that the skin regenerating properties of Pepsensyal contributes to smoothing of the crow's feet area, and to reduce the appearance of wrinkles, with a statistically significant difference *vs* the placebo after only 14 days of use.

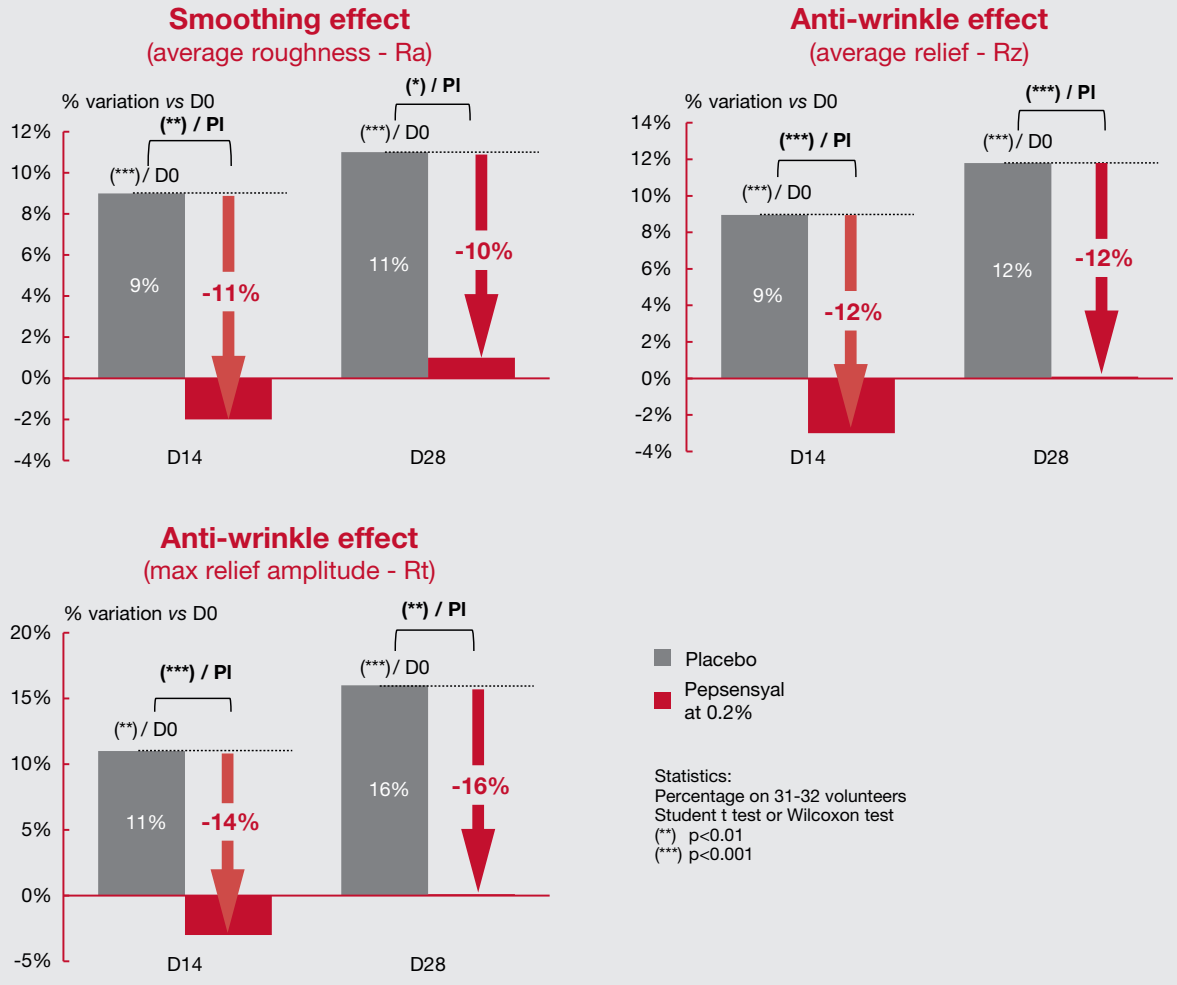


Figure 16 - Mean percentage of change of average roughness (Ra), average relief (Rz) and maximum relief amplitude (Rt) on the crow's feet area after application of the 0.2% Pepsensyal serum *vs* the placebo serum.

in vivo

Pictures in figures 17 and 18 illustrate this smoothing and anti-wrinkle effect with Pepsensyal, overtime and comparatively to placebo.

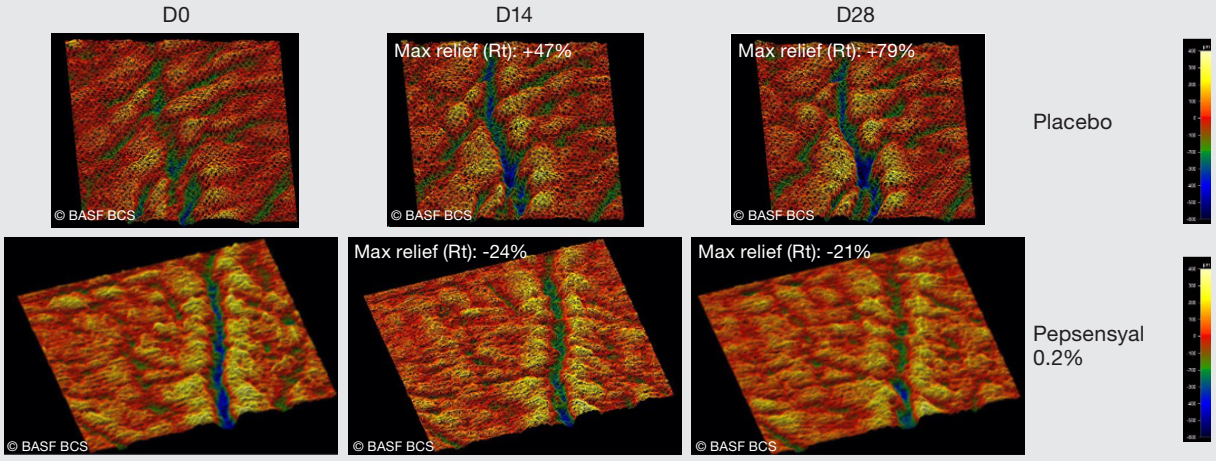


Figure 17 - Illustrative pictures of the fringe projection measurements with Pepsensyal at 0.2% and placebo, at D0 (baseline, before treatment), and after 14 and 28 days of product use.

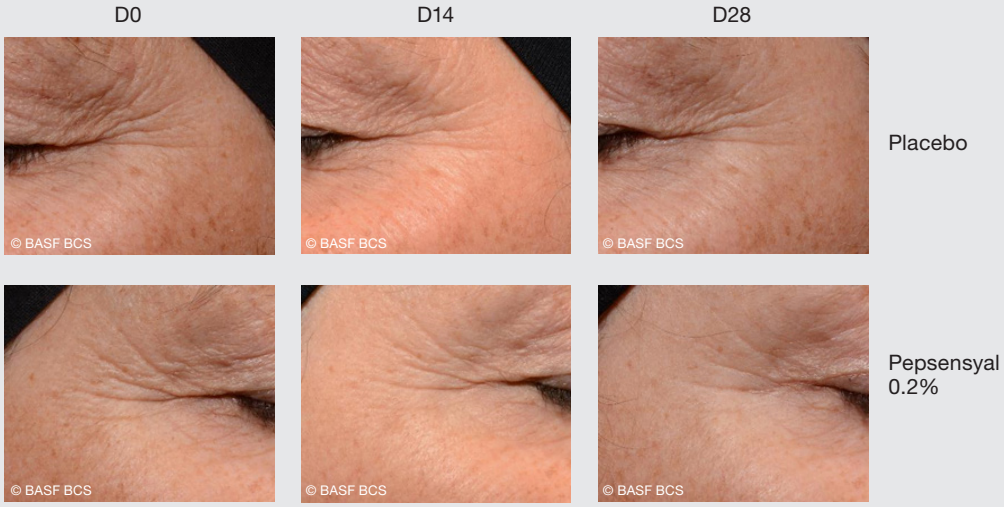


Figure 18 - Illustrative pictures of the appearance of crow's feet wrinkles with Pepsensyal at 0.2% and placebo, at D0 (baseline, before treatment), and after 14 and 28 days of product use.

CONCLUSION

Pepsensyal at 0.2% improved the signs of skin aging with a re-densified dermis, a skin surface restructured and a reduced appearance of wrinkles. After only 14 days or 28 days of use, dermis density was increased, skin surface relief isotropy was improved, and crow's feet wrinkles appeared smoother. Pepsensyal thus supports skin regeneration, helping to slow-down the visible signs of aging for a youthful skin appearance.

MATERIALS&METHODS

Study design

The clinical study was carried-out randomized, double-blind, and on split-face *vs* placebo under dermatological control. The efficacy of the serum containing Pepsensyal at 0.2% was compared to the baseline (before treatment, D0) and to the placebo serum (same formulation without Pepsensyal). The study was conducted during a period of 28 days with check points at D0, D14 and D28.

Inclusion criteria

The study was done on 32 healthy female volunteers with Fitzpatrick skin type I, II, III or IV, aged from 44 to 64 years old (mean 54.7) with dry skin on the face, wrinkles and fine lines in the Crow's feet, and loss of firmness on the lower face.

Application modality

Volunteers applied twice a day, once in the morning and once in the evening, for 28 days the Pepsensyal serum at 0.2% or the placebo serum on half face, under normal conditions of use. The last application was done the evening before the visit.

Evaluation methods

Dermis density measurement by ultrasound echogenicity on the lower face and Image analysis

Measurements were realized directly *in vivo* on the lower face, using the high frequency echograph Dermascan C 2D. The measurement principle is that of the ultrasound scanner: an ultrasound beam is emitted by a piezoelectric ceramic. This beam is partially reflected by the interface separating two media of different ultrasound impedance. The ultrasound used has a 20 MHz probe. The probe was applied directly to the skin. A contact gel provided homogeneous diffusion of the signal. This method allows the bi-dimensional visualization of the skin, on the epidermis and dermis levels, with a 13 mm penetration. During the picture acquisition, the conjunctive tissue, echogenic, appears in color (from yellow to green); non-echogenic tissues (such as water) appear in black. The ratio of the non-echogenic surface to the total surface analyzed gives the proportion of non-echogenic tissues (in %). A decrease in this non-echogenic ratio, or increase of echogenic ratio, characterizes an increase in the dermis density. Visually, a decrease in black pixels characterizes an increase in the dermis density.

Fringe projection: skin roughness, wrinkles on crow's feet area and skin surface isotropy on the cheeks

Measurements were done directly *in vivo* on crow's feet area for skin roughness and wrinkles, and on the cheek for skin surface isotropy, using the PRIMOS Lite system (Phaseshift Rapid *In vivo* Measurement Of Skin). This technique consists in calculating a phase image from images with interference fringe projection. This image then allows to determine the height of each point. The acquisition software allows to obtain 2D and 3D measurements and to determine parameters of the cutaneous relief on 32 radiuses distributed like a star on the zone of interest. An automatic system of repositioning allows the precise re-identification of the zone of measurement.

On crow's feet area, measured parameters were:

Ra = average roughness. It is the arithmetic average of the absolute values (not differentiating the peaks and troughs) of the heights along the length of the profile (Fig. 19). A decrease of Ra characterizes a smoothing effect.

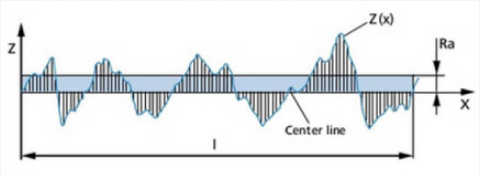


Figure 19 - Average roughness Ra.

Rz = average maximum height = average relief. It is the average of five maxima on the profile length (ln). A maximum corresponds to the difference between the highest peak and the deepest trough within a measuring section (lr): Rz= (Rz1+Rz2+Rz3+Rz4+Rz5)/5 (Fig. 20). A decrease of Rz characterizes an anti-wrinkles effect.

Rt = maximum profile height = maximum amplitude of the relief. It is the difference in height between the highest peak and the deepest trough within the total measuring section (ln) (Fig. 20). A decrease of Rt characterizes an anti-wrinkles effect.

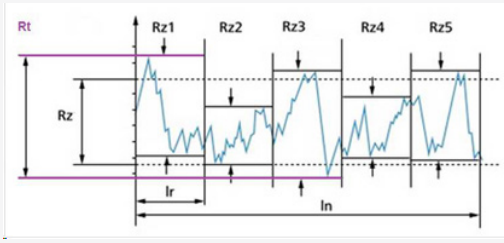


Figure 20 - Average maximum height Rz and maximum profile height Rt.

On the cheeks, the measured parameter was the isotropy. It defines the orientation of the lines of the cutaneous relief. The aging mechanism results in a change in the organization of skin lines passing of a homogeneous isotropic state (lines oriented in all directions) for a young person to a state where small furrows gradually disappear to give rise to a state where only persist deep furrows (privileged directions). An increase in the percentage of isotropy characterizes a skin restructuring and anti-aging effect of the studied product.

Results and statistics

Results are expressed as the mean percentage of change compared to the baseline measurement (D0). The statistical analysis of the evolution of the parameters as a function of time has been done after the verification of the normality of distribution using Shapiro-Wilk test. The statistical analysis of the change in the studied parameters for each product over time as well as the differences in the studied parameters between the treatment groups were performed as follows:

- validation of the normality of the studied parameters: paired Student t test,
- invalidation of the normality of the studied parameters: Wilcoxon test.



GENERAL CONCLUSION

Age slowing peptide

- Responds to growing consumer trend for slow and healthy aging.
- Inspired by a human naturally occurring tetrapeptide & the natural skin regeneration process.
- Contributes *in vitro* to slow cell-aging, epidermis and dermis regeneration.
- Clinically proven to re-densify skin, restructure the skin's surface, and smooth the appearance of crow's-feet wrinkles.

Consumers have an increased desire for cosmetic responding to the slow-aging trend. This movement reflects a shift in consumer attitudes: rather than fighting the natural aging process, consumers are increasingly concerned with healthy aging through a holistic, preventative approach to skin care. Inspired by a human naturally occurring peptide and by skin regeneration process, Pepsensyal is a relevant and science-backed ingredients to answer to this need.

In vitro evaluation of performance evidenced for Pepsensyal epidermal and dermal regeneration properties at nanomolar concentrations. It was shown to boost expression by keratinocytes and fibroblasts of the cell longevity protein Sirtuin-1, and its enzymatic deacetylase activity. It upregulated the growth of human epidermal keratinocyte stem and progenitor cells, and the synthesis of the tight junction proteins ZO-1, claudin-1, and occludin. *Ex vivo* on skin explants, it upregulated some structural components of the dermo epidermal junction and dermal extracellular matrix: glycosaminoglycans, collagen III, collagen IV and fibronectin.

Tested *in vivo* for 28 days in a placebo-controlled clinical study on 32 volunteers, Pepsensyal at 0.2% demonstrated after only 14 days an increase of skin isotropy by 15% *vs* placebo, indicating a restructuring effect on the skin surface. At the same time crow's feet wrinkles appeared smoothed with a 14% decrease in maximum wrinkle amplitude *vs* placebo. After 28 days, the dermis was also re-densified with a 21% increase *vs* placebo.

Pepsensyal (INCI: Mannitol (and) Acetyl sh-Tetrapeptide-1 (and) Lauroyl sh-Tetrapeptide-1) is a tetrapeptide promoting skin regeneration for a re-densified skin with smoothed wrinkle appearance, meeting the consumer quest for slow and healthy aging.

ANNEX

Annex 1 - Technical data - Available upon request

- Quality and regulatory product information
- Composition data sheet
- Specifications
- Product Purity Profile
- Formulation Data Sheet
- Natural content in BASF Personal Care Active product EMC

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